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COURSE NAME: FULL STACK WEB DEVELOPMENT

CODE: 23CSM4121

TITLE OF THE EXPERIMENT: To create a Student Profile Manager using JavaScript Class and DOM

DESCRIPTION:

The Student Profile application is a dynamic webpage developed using **HTML, CSS, and JavaScript**. It allows the user to enter the student's **Name, Roll Number, Department, and CGPA** through input fields. A JavaScript **Student class** is created with these properties, and an object is created using the values provided by the user.

When the **Create Profile** button is clicked, JavaScript retrieves the entered values using DOM methods and creates a Student object. The student details are then dynamically displayed on the webpage using **DOM manipulation**, `createElement()`, `textContent`, and `appendChild()`. Basic CSS styling is applied to make the generated student profile visually clear and attractive.

This program demonstrates the use of **JavaScript classes and objects, DOM selection, dynamic element creation, content modification, adding elements to the webpage, and event handling**

PROGRAM:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Student Profile</title>
```

```
<style>
```

```
  body {  
    font-family: Arial;  
    background-color: #f2f2f2;  
    text-align: center;  
    padding: 30px;  
  }
```

```
  input {  
    padding: 10px;  
    margin: 5px;  
    width: 250px;  
  }
```

```
  button {  
    padding: 10px 20px;  
    background-color: blue;  
    color: white;  
    border: none;  
    cursor: pointer;  
    margin-top: 10px;  
  }
```

```
  #profile {
```

```
background-color: white;
width: 350px;
margin: 25px auto;
padding: 20px;
border-radius: 10px;
box-shadow: 0 0 10px gray;
text-align: left;
}
```

```
#profile h2 {
  text-align: center;
  color: blue;
}
```

```
#profile p {
  font-size: 16px;
  margin: 12px;
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<h1>Student Profile</h1>
```

```
<input type="text" id="name" placeholder="Enter Name"><br>
```

```
<input type="text" id="roll" placeholder="Enter Roll Number"><br>
```

```
<input type="text" id="dept" placeholder="Enter Department"><br>
```

```
<input type="text" id="cgpa" placeholder="Enter CGPA"><br>
```

```
<button onclick="createProfile()">Create Profile</button>
```

```
<div id="profile"></div>
```

```
<script>
```

```
// Student Class
```

```
class Student {
```

```
    constructor(name, roll, department, cgpa) {
```

```
        this.name = name;
```

```
        this.roll = roll;
```

```
        this.department = department;
```

```
        this.cgpa = cgpa;
```

```
    }
```

```
}
```

```
// Function to create student profile
```

```
function createProfile() {
```

```
    // Get values from input fields
```

```
    let name = document.getElementById("name").value;
```

```
    let roll = document.getElementById("roll").value;
```

```
let dept = document.getElementById("dept").value;
let cgpa = document.getElementById("cgpa").value;

// Create Student object
let student = new Student(name, roll, dept, cgpa);

// Select profile div
let profile = document.getElementById("profile");

// Clear previous profile
profile.innerHTML = "";

// Create elements dynamically
let heading = document.createElement("h2");
heading.textContent = "Student Profile";

let namePara = document.createElement("p");
namePara.textContent = "Name      : " + student.name;

let rollPara = document.createElement("p");
rollPara.textContent = "Roll No   : " + student.roll;

let deptPara = document.createElement("p");
deptPara.textContent = "Department : " + student.department;

let cgpaPara = document.createElement("p");
```

```
cgpaPara.textContent = "CGPA      : " + student.cgpa;
```

```
// Add elements to webpage
```

```
profile.appendChild(heading);
```

```
profile.appendChild(namePara);
```

```
profile.appendChild(rollPara);
```

```
profile.appendChild(deptPara);
```

```
profile.appendChild(cgpaPara);
```

```
}
```

```
</script>
```

```
</body>
```

```
</html>
```

OUTPUT:

The screenshot displays a web application interface for creating a student profile. It features a form with four input fields and a 'Create Profile' button. Below the form, a preview box shows the rendered HTML output, which includes a heading and four paragraphs of student information.

Student Profile

Ravi Kumar

A231265520001

CSE(AI&ML)

8.42

Create Profile

Student Profile

Name : Ravi Kumar

Roll No : A231265520001

Department : CSE(AI&ML)

CGPA : 8.42